DONUT: Design and Optimization of Non-axisymmetric Unified Tori

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Abstract

DONUT is a a computational geometry tool for optimising stellarator-type fusion reactors. It generates toroidal, non-axisymmetric plasma surfaces and coil geometries, enabling researchers to design and optimize complex 3D magnetic confinement systems.

Nomenclature

α	Twist parameter
$\bar{\delta}$	Average triangularity
$\bar{\iota}$	Edge rotational transform over number of field periods
ϵ_{max}	Maximum elongation
ϕ	Toroidal angle
ϕ_i	azimuthal angle
ϕ_t	Toroidal spherical coordinate polar angle
ψ_p	Poloidal circular coordinate angle
$\mathrm{N_p}$	Number of poloidal sections
$\mathrm{N_t}$	Number of toroidal sections
θ	Poloidal angle
θ_i	polar angle
θ_t	Toroidal spherical coordinate angle
A	Aspect ratio
m	knot count relationship
n	number of control points
$N_{i,q}(u)$	B-spline basis functions of degree q
P_i	NURBS control points
P_m	Modulation period
r_i	radial distance
r_p	Poloidal circular coordinate radius
r_t	Toroidal spherical coordinate azimuthal radius
s_t	Parametric poloidal section location
$u_0, u_1, ...u_m$	knot vector

w_i	NURBS associated weights
w_p	Poloidal NURBS associated weights
w_t	Toroidal NURBS weight

1 Introduction

The optimisation of stellarator plasma geometry is a central challenge in modern magnetic confinement fusion research. Unlike tokamaks, which rely partly on plasma current to generate rotational transform, stellarators achieve plasma confinement entirely through externally generated three-dimensional magnetic fields [18]. This intrinsic steady-state capability makes stellarators attractive candidates for future fusion reactors, particularly because they avoid many of the current-driven instabilities[4] [14] [7] [13] [1] [17] [8] and disruption risks [12] [6] [3] [15] [10] [9] [16] associated with tokamaks. However, the absence of axisymmetry introduces substantial geometric complexity, requiring careful optimisation of magnetic surfaces, coil configurations, and plasma equilibrium properties.

In stellarator design, plasma geometry strongly influences confinement quality, neoclassical transport, magnetohydrodynamic (MHD) stability, energetic particle confinement, and engineering feasibility. The shape of the plasma boundary determines the structure of magnetic field lines and the resulting particle trajectories, making geometric optimisation a multidimensional problem involving both physics performance and practical constraints. Modern stellarator optimisation therefore seeks configurations that simultaneously minimise transport losses, improve stability, reduce bootstrap current, and maintain realizable coil geometries.

[5] is a recent example of a stellarator optimisation code that provides a dataset of optimized or partially optimized stellarator plasma boundary configurations that exhibit quasi-isodynamic (QI)-like properties. It defines the geometry using a Fourier surface representation in cylindrical coordinates, where the boundary shape is parameterized by Fourier coefficients for the radial $R(\theta, \phi)$ and vertical $Z(\theta, \phi)$ components. The current work targets the multi-objective constrained optimisation problem, in particular the Geometry problem, by using an alternative plasma boundary representation based on Non-Uniform Rational B-Splines (NURBS). This approach allows for higher geometric expressivity and has the natural ability to enforce constriants. It is also easier to enocode Computer Aided Design (CAD) logic, enabling seamless integration with engineering design workflows.

In future work, this geometry will be used to define the computational domain for a GPU-accelerated MHD equilibrium solver that operates in physical space rather than flux coordinates to solve for MHD equilibrium. Minimum normalised magnetic gradient scale length can then be used for Simple-to-build QI and MHD-stable-QI type problems. The source code of this work is available at `https://github.com/DjentleViBe/DONUT`.

2 Geometry

The Geometric problem is originally defined as:

$$\min_{\theta} \epsilon_{\text{max}} \tag{1}$$

$$\text{s.t. } A \leq A^*, \tag{2}$$

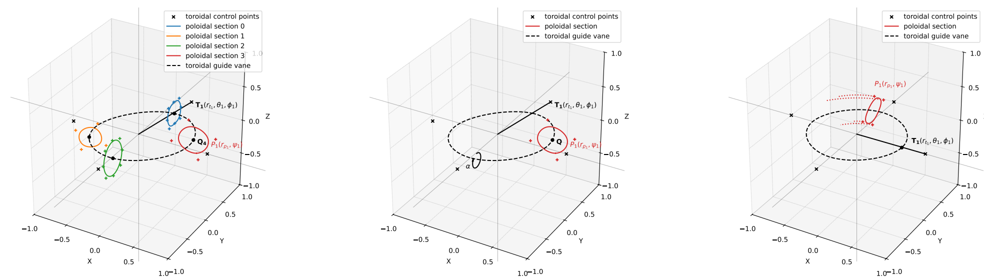
$$\bar{\delta} \leq \bar{\delta}^*, \tag{3}$$

$$\bar{\iota} \leq \bar{\iota}^* \tag{4}$$

A plasma boundary is defined by the coefficients R_{mn} and Z_{mn} of a truncated Fourier series in cylindrical coordinates, parameterized by θ and ϕ given in Appendix A

In this paper, the plasma boundary is defined instead by first constructing a 3D NURBS curve from points expressed in spherical coordinates. This serves as the basis for the toroidal path. Using this basis curve, the toroidal geometry can be constructed using three different approaches:

- **Type 1** - A prescribed number of poloidal cross-sections are then generated, each represented as a closed 2D NURBS loop. Finally, these sections are lofted along the 3D guiding curve,



(a) (b) (c)

Figure 1: (a) Type 1 geometry generated with four poloidal cross section $(Q_1, ..Q_4)$ definitions and four 3D control points $(T_1, ..T_4)$ in spherical coordinate system for the toroidal guide vane. The poloidal section at Q_4 consists of four 2D control points in circular coordinate system. Each poloidal section is moved by rotating along the guide vane with its orientation perperndicular to the tangent at that guide vane point. Only Q_4 and T_1 are depicted. (b) Type 2 geoemtry with poloidal section lofted along the 3D toroidal spline with twist parameter α defined along the 3D guide vane. (c) Type 3 geometry with spatially varying P(r_p, ψ) along the toroidal guide vane.

with the assumption of tangentiality at each poloidal section, to form the complete geometry.

- **Type 2** - A single poloidal base cross-section is defined using a closed 2D NURBS curve. A twist parameter is then specified along the 3D guide curve. The base cross-section is lofted along the guide vane, while twist and scaling transformations are applied about the local tangential direction.
$$\alpha(t) = k \cdot t \cdot \frac{2\pi}{N} + B \cdot \sin \frac{N \cdot t \cdot 2\pi}{P_m} \tag{5}$$
The sensitivity of α is given in Appendix C.
- **Type 3** - The r_p and ψ parameters defining the 2D spline are treated as spatially varying functions along the guide vane.
- **Type 4** - A candidate geometry is represented as the union of N spherical primitives, each defined by its centre coordinates and radius (x_i, y_i, z_i, r_i). The parameters of all spheres form the chromosome used by the genetic algorithm. Geometric variation is achieved by modifying sphere positions and radii, allowing both shape and topology changes during optimisation. Compared with voxel-based encodings, sphere-based representations provide a more compact design space while retaining the ability to represent complex three-dimensional geometries. Similar primitive-based approaches have been used in evolutionary shape optimisation and explicit topology optimisation frameworks such as those of [2] [11].

3 Optimisation

A NURBS curve of degree p is defined as:

$$\mathbf{C}(u) = \frac{\sum_{i=0}^{n} N_{i,q}(u)\, w_i\, \mathbf{P}_i}{\sum_{i=0}^{n} N_{i,q}(u)\, w_i}, \qquad u \in [u_0, u_m]. \tag{6}$$

The control points $\mathbf{P}_i$ are defined in spherical coordinates and transformed into Cartesian space as

$$\mathbf{P}_i = \begin{bmatrix} r_i \sin\theta_i \cos\phi_i \\ r_i \sin\theta_i \sin\phi_i \\ r_i \cos\theta_i \end{bmatrix},$$

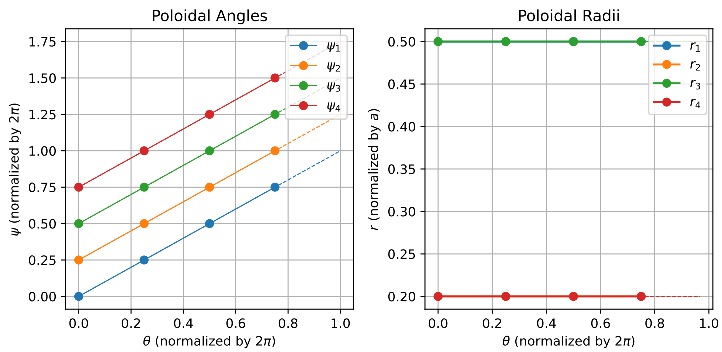


Figure 2: Fitting NURBS curve for ψ and r values of Type 3 geometry. Periodic representations require the last point to be adjusted accordingly.

3.1 Gradient based methods

The following variables are parametrisable: $(\theta_t, \phi_t, r_t, w_t)_{\mathrm{N_t}}$, $(s_t, \psi_p, r_p, w_p)_{\mathrm{N_p}}$. The objective function from Equation 1 is used along with constraint equations 2 and 3 for the non-linear optimisation problem. Equation 4 is irrelevant as rotational edge transform is naturally encoded in the poloidal section. Solutions obtained using `Scipy` algorithms `SLSQP`, `L-BFGS-G`, `COBYLA`, `COBYQA`, `Nelder-Mead`, `trust-constr` and `Powell` are provided. For `trust-constr` and `SLSQP`, constraints are modelled as Non-Linear representations while for others it is included as a penalty term in the objective function.

3.2 Genetic Algorithm

The genetic algorithm was implemented using the Type 2 geometry parameterization. Each individual in the population was represented by a chromosome containing the complete set of normalized geometry parameters, including toroidal radii, poloidal radii, angular coordinates, section spacing parameters, and global shaping parameters. Each parameter within the chromosome constituted a gene that defined a specific aspect of the geometry.

The initial population was generated using Latin Hypercube Sampling (LHS), which provides a more uniform coverage of the design space than purely random sampling. Objective history is given in Appendix B

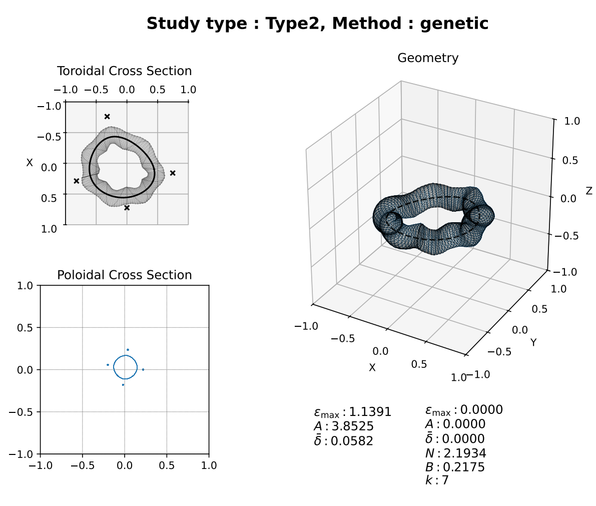


Figure 3: Results for Type 2 geometry after 100 generations and an initial population size of 300.

Table 1: Maximum elongation across different geometry types after optimization using Gradient based methods.

	Type 1		Type 2		Type 3	
	ϵ_{init} = 4.8509		ϵ_{init} = 1.4985		ϵ_{init} = 2.6161	
Optimiser	`nfev`	$\epsilon_{\max}$	`nfev`	$\epsilon_{\max}$	`nfev`	$\epsilon_{\max}$
L-BFGS-B	372	3.0968	672	1.3620	69	2.6161
Nelder-Mead	95	4.3647	88	1.3766	70	2.6152
Trust-constr	186	4.7543	64	1.4617	138	1.4992
COBYLA	184	2.7906	62	1.1257	136	1.7699
COBYQA	188	3.7076	66	1.2460	140	2.6151
SLSQP	283	3.5246	101	1.1341		
Powell	184	4.0724	62	1.4957	136	2.6156

3.3 Neural Networks

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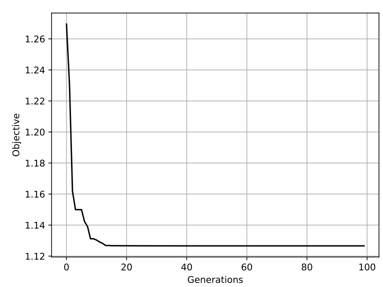
A Fourier Geometry

$$R(\theta,\phi) = \sum_{n=-N}^{N} \sum_{m=0}^{M} R_{mn} \cos(m\theta - n\mathrm{N}_{\mathrm{fp}}\phi) \tag{7}$$

$$Z(\theta,\phi) = \sum_{n=-N}^{N} \sum_{m=0}^{M} Z_{mn} \sin(m\theta - n\mathrm{N}_{\mathrm{fp}}\phi) \tag{8}$$

B Generations



C Twist parameter